

Adapting Vision–Language Models for Generalizable Anomaly Detection in Medical Images

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Abstract

Medical anomaly detection is an important task in computer-aided diagnosis, yet training robust models is difficult due to limited labeled medical data and strong domain differences between natural and medical images. In this project, we implement and extend the Multi-level Visual Feature Adaptation (MVFA) [1] framework for few-shot medical anomaly detection. Using the RSNA Pneumonia Detection dataset, we evaluate several improvements, including adapter regularization, medical prompt engineering, Low-Rank Adaptation (LoRA) [2], and a learnable Multi-Layer Perceptron (MLP) anomaly scoring head. Experimental results show that strong regularization improves stability under extremely low-shot regimes, while direct visual anomaly regression using an MLP achieves the best performance when more support samples are available.

1. Introduction

1.1 Motivation

Medical anomaly detection plays a crucial role in clinical diagnosis and early disease identification. Automatically identifying abnormal patterns in medical images can help clinicians detect diseases such as pneumonia, tumors, and other pathological conditions. However, training reliable models for medical anomaly detection remains challenging because labeled medical datasets are often scarce and expensive to obtain.

Traditional deep learning approaches typically require large labeled datasets and may struggle to generalize when training data is limited. Recent advances in vision–language models (VLMs), such as CLIP, have demonstrated strong generalization by learning joint representations of images and text from large-scale datasets [3]. These models enable zero-shot and few-shot learning, allowing models to recognize concepts with minimal labeled examples. Despite their success, applying CLIP directly to medical imaging tasks presents a significant challenge. CLIP is primarily trained on natural image–text pairs, and consequently, its learned representations do not directly align with the visual characteristics of medical images [4]. This creates a domain gap that can reduce anomaly detection performance. To address this issue, recent work has proposed adapting vision–language models for medical anomaly detection through lightweight feature adaptation techniques. In particular, the Multi-level Visual Feature Adaptation (MVFA) framework introduces residual adapters that modify CLIP’s visual features at multiple levels, thereby enabling better alignment between visual representations and medical anomaly-detection objectives [1].

1.2 Objective

The objective of this project is to evaluate and improve the MVFA-based framework for few-shot medical anomaly detection. Our experiments are conducted using the RSNA Pneumonia Detection Challenge dataset, where anomaly detection performance is evaluated using the AUROC metric across several few-shot settings.

To improve the baseline MVFA architecture, we explore several modifications that aim to enhance model adaptability under limited supervision. These include adapter regularization to stabilize training, medical-domain prompt engineering, LoRA[2] injected into the visual backbone, and a learnable Multi-Layer Perceptron (MLP) anomaly-scoring head that predicts anomaly scores directly from visual embeddings.

The main contributions of this project are summarized as follows:

- Implementation and evaluation of the MVFA anomaly detection framework for medical imaging tasks.
- Investigation of architectural improvements, including adapter regularization, medical prompt engineering, LoRA backbone adaptation, and MLP-based anomaly score prediction.
- Development of a multi-score anomaly detection strategy combining cosine similarity, Mahalanobis distance, and energy-based scoring.
- Comprehensive evaluation across multiple few-shot regimes, demonstrating how different architectural choices affect anomaly detection performance under limited data.
- Evaluation of BiomedCLIP as an alternative, a domain-specific variant of CLIP trained on biomedical image-text pairs [5].

2. System Design and Implementation

2.1 Algorithms

The project builds upon the MVFA-AD framework, which leverages multimodal representation learning to detect anomalies under zero-shot and few-shot settings. The method relies on a vision–language model backbone (CLIP) to encode medical images and textual prompts into a shared embedding space, enabling anomaly detection by comparing image features with descriptive prompts for normal and abnormal conditions. The baseline system follows the original MVFA-AD architecture, in which the CLIP visual encoder produces image embeddings aligned with text embeddings representing normal and abnormal states. Anomaly scores are derived from the similarity between these visual and textual representations. In this project, the baseline implementation serves as the reference model against which all architectural modifications and improvements are evaluated. To improve adaptability under limited supervision, LoRA layers are injected into the visual backbone, allowing parameter-efficient fine-tuning without retraining the full model. The system also incorporates domain-specific prompt engineering, where medical prompts tailored to pneumonia-related abnormalities guide the model toward clinically meaningful representations.

As an alternative approach, we experimented with removing the reliance on text features entirely, replacing the similarity-based scoring mechanism with an MLP that directly takes the learned image feature representation as input and predicts an anomaly score. This modification eliminates reliance on text embeddings while allowing the model to learn a direct mapping from visual features to anomaly likelihood, thereby improving performance in our experiments. During training, the anomaly scoring mechanisms are optimized to distinguish normal and abnormal samples within the few-shot support set. In configurations that use the MLP anomaly scorer, the model learns a direct regression-style mapping from visual embeddings to anomaly likelihood scores, which are optimized using a binary classification objective.

2.2 Technologies and Tools

We used the NVIDIA V100 GPUs on the WAVE HPC and NVIDIA RTX 5070Ti to evaluate our methodologies. Our implementation was built on PyTorch, and the core vision-language backbone was OpenAI's CLIP (ViT-L/14). We also experimented with Microsoft BiomedCLIP, a domain-adapted alternative given its pretraining on large-scale biomedical image-text pairs. Metrics computation was handled using Scikit-Learn, which provided robust implementations of AUROC calculation and logistic regression for feature fusion. For dataset analytics and result interpretation, we employed Matplotlib and Seaborn to build custom visualization pipelines that surfaced distribution-level insights across few-shot settings and anomaly categories.

2.3 Network Architecture

Several network architectures were explored to improve upon the baseline performance of the MVFA-AD anomaly detection framework. While the original method relies on multimodal feature alignment between image and text representations, this project investigates architectural variations that may better capture abnormal patterns in medical images. Different backbone encoders, feature fusion strategies, and lightweight adaptation mechanisms were evaluated to determine whether they could enhance anomaly discrimination under limited supervision. The diagram shows the proposed architecture for anomaly

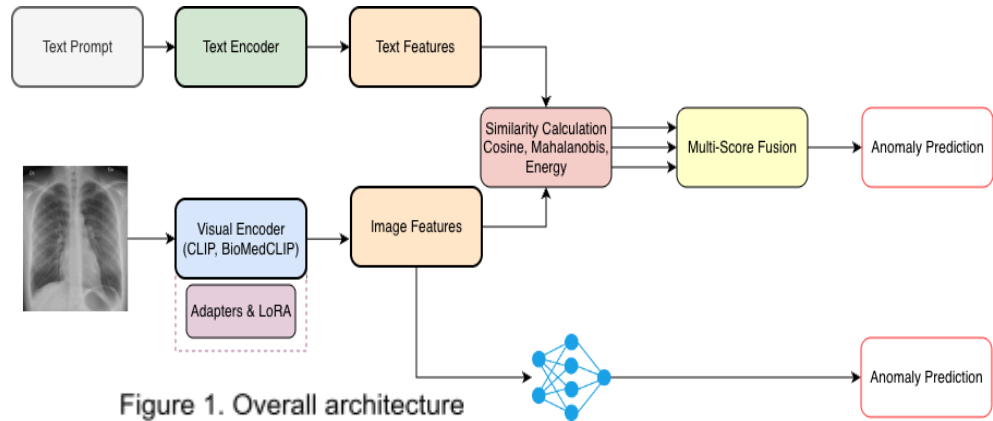


Figure 1. Overall architecture

detection. The system processes chest X-ray images using a CLIP-based visual encoder enhanced with adapters and LoRA modules. Anomaly scores are computed either through similarity with medical text prompts or through an MLP anomaly scorer, and final predictions are obtained using a multi-score fusion mechanism.

2.3.1 Anomaly Score Prediction Architecture

To explore an alternative mechanism for anomaly scoring, we replaced the original interaction between text and image features with a lightweight MLP that predicts the anomaly score directly from the image feature representation. Instead of computing similarity between visual and textual embeddings, the model relies solely on the semantic information contained in the visual encoder output. This modification removes the dependency on text prompts and allows the network to learn a direct mapping between the image embedding space and the anomaly score. The MLP's input is the 768-dimensional image feature vector produced by the visual encoder. The MLP consists of two fully connected layers. The first layer projects the input features from 768 dimensions to a hidden representation of 256 dimensions, followed by a GELU activation function, which introduces non-linearity and helps the network capture complex relationships between image features and abnormal patterns. The second layer maps the 256-dimensional hidden representation to the final anomaly score, and no activation function is applied to this output layer, allowing the model to produce an unrestricted continuous score suitable for anomaly detection. This architecture provides a simple yet effective alternative to multimodal similarity-based scoring. By learning a direct regression-style mapping from image features to anomaly scores, the MLP can adaptively identify patterns associated with abnormal samples without relying on predefined textual descriptions, thereby improving performance in practice.

2.3.2 Adapter Regularization Architecture

Our core adapter optimization involved a significant architectural shift towards strict regularization to prevent catastrophic overfitting on N-shot support sets. While the original framework mapped visual features to the target domain, it suffered from severe instability under extremely low-shot data constraints (2-shot or 4-shot). To counteract this, we implemented a weight-decay mechanism ($1e-4$) applied specifically to the adapter's dense linear layers, alongside a gradient clipping filter with a maximum norm of 1.0. This architectural guardrail ensures that the adapter does not shift the patch embeddings so far that they lose the semantic grounding of the original CLIP space, effectively stabilizing training dynamics when reference images are scarce.

2.3.3 Low-Rank Adaptation (LoRA) Integration

To explore adaptation mechanisms beyond the standard adapter network, we bypassed the conventional approach by directly injecting LoRA matrices into the frozen Vision Transformer (ViT) backbone. Specifically, we targeted the in-projection and

out-projection weights of the final Transformer blocks (blocks 21, 22, and 23 for our high-capacity configurations). By embedding these learnable, low-rank matrices ($r=8$, $\alpha=8$), we allowed the model to adapt its internal visual attention mechanisms to emphasize medical morphologies. This method provides a parameter-efficient pathway to tune the core visual representations without incurring the massive computational overhead or catastrophic forgetting associated with full fine-tuning.

2.3.4 Multi-Score Fusion Architecture

Instead of relying on a single distance metric for anomaly scoring, our advanced architectures employ a sophisticated 3-score fusion head to comprehensively evaluate target images. This architecture simultaneously computes three independent metrics: cosine similarity, which compares global visual features to ensembled medical text prompts; Mahalanobis distance, which fits a Gaussian distribution to the normal support features to detect spatial outliers; and energy scoring, which calculates the logsumexp of the logits to estimate the inherent "unfamiliarity" of a visual patch. To ensure optimal combination, these three distinct scores are dynamically weighted and calibrated during inference using a logistic regression model trained strictly on the isolated k-shot support set. During inference, each test image is passed through the frozen visual encoder with the inserted adaptation modules. The resulting feature embeddings are evaluated using the different anomaly scoring mechanisms, and the calibrated fusion model combines the scores into a final anomaly prediction.

2.4 Hyperparameters

The model was trained using a set of standard hyperparameters selected to balance training stability and computational efficiency. Training was conducted for 50 epochs, although empirical observations showed that the model typically converged much earlier, between approximately 2 and 12 epochs, indicating that the learning dynamics stabilize quickly in the few-shot regime. A learning rate of $1e-3$ paired with a cosine annealing scheduler was used, providing a stable optimization process while allowing the model to update parameters effectively during early training. Additionally, we structurally modified the inference pipeline to support a batch size of 32, a critical optimization over the baseline batch size of 1, which maintained a reasonable trade-off between gradient stability and GPU memory usage while rendering our massive 16-experiment progression physically viable. For our specialized adaptations, LoRA configurations used $r=8$ and $\alpha=8$, while the regularization branches enforced weight decay of $1e-4$ and gradient clipping at 1.0. Furthermore, to optimize classification margins across discrete semantic boundaries, the inference temperature was systematically swept over the intervals $\tau \in [0.7, 1.0, 1.3, 1.6]$. Together, these hyperparameters provided reliable convergence and consistent training behavior across the diverse combinatorial experiments.

3 Experiments and Evaluation

3.1 Dataset

We use the RSNA chest X-ray dataset from Kaggle, which consists of frontal chest radiographs annotated into three categories: Normal, Pneumonia, and Not Normal / Not Pneumonia. Following the same preprocessing protocol as the original paper, we merge the Pneumonia and Not Normal/Not Pneumonia classes into a single Abnormal label. This labeling strategy reframes the task as a binary classification problem, focusing on distinguishing normal cases from any abnormal findings. The original dataset is structured as follows:

- Image resolution: 1024 x 1024
- Size: 6GB
- Number of images: ~29,000 samples
- Non-zero requirements: Each image has $1024 \times 1024 > 1$ million pixels

After consolidating the Pneumonia and Not Normal / Not Pneumonia classes into a single abnormal category, we applied additional preprocessing and filtering to create our final benchmark. Ambiguous or poorly labeled cases from the original RSNA training batch were removed to improve label reliability and reduce noise in evaluation.

The final preprocessed dataset used in our experiments contains 17,194 total images, with a class distribution of 5,881 Normal cases and 11,313 Abnormal cases. This filtered benchmark was then used for all few-shot and zero-shot experiments in the report. To match the expected CLIP input format, grayscale chest X-ray images were converted to RGB. All images were resized to 224×224 to match the model’s input resolution. Finally, pixel values were standardized to have a zero mean and unit variance, ensuring stable, consistent model behavior during training and evaluation.

3.2 Methodology

Our evaluation protocol follows a strict k-shot stratified sampling procedure designed to ensure fair and reproducible comparisons across all experimental configurations. For each N-shot setting ($N \in \{2, 4, 8, 16\}$), we construct the adapter training set by sampling exactly N normal and N abnormal images, which results in a balanced support set of (2N) samples. This balanced sampling is critical in the few-shot regime, where class imbalance can disproportionately bias model learning and artificially inflate performance. All remaining images from the filtered benchmark are held out and used exclusively for evaluation, ensuring that no information from the test set leaks into training. Under this protocol, the support set sizes are 4, 8, 16, and 32 images for the 2-shot, 4-shot, 8-shot, and 16-shot settings, respectively, while the remaining images are used for testing. This protocol mirrors the original MVFA-AD evaluation methodology and allows us to directly measure how well the model generalizes from extremely limited supervision. In the zero-shot setting, no labeled support images from the target dataset are used. Predictions are instead generated directly from the pretrained visual-language alignment and the associated medical prompt representations. This setting evaluates how well the model distinguishes normal and abnormal chest X-ray images without any task-specific adaptation to target support samples.

Model performance is evaluated using the Area Under the Receiver Operating Characteristic Curve (AUROC). AUROC is particularly appropriate for anomaly detection tasks because it measures the model’s ability to rank abnormal samples higher than normal samples across all possible decision thresholds, rather than relying on a single fixed classification threshold [1]. This makes the metric robust to variations in score calibration and provides a threshold-independent assessment of detection capability. In the context of the RSNA Pneumonia Detection dataset, AUROC captures how effectively the model distinguishes pathological chest X-ray images from normal cases under very limited training data. To rigorously quantify the contribution of each proposed modification, each experimental variant is directly compared with the unmodified MVFA-AD baseline under identical few-shot conditions. This controlled comparison isolates the effect of each architectural or algorithmic change, such as prompt engineering, backbone replacement, LoRA adaptation, or the learnable anomaly score prediction MLP, without confounding factors from differing data splits or training protocols. Each modification is evaluated across all four shot settings (2, 4, 8, and 16), enabling us to observe not only overall performance improvements but also how each technique scales with increasing supervision. This systematic evaluation framework ensures that any observed improvements can be confidently attributed to the proposed changes rather than experimental variance.

3.3 Results and Analysis

3.3.1 Experimental Setup

To systematically evaluate the effectiveness of the proposed architectural modifications, we designed sixteen experimental configurations. These experiments were structured to isolate the contribution of different components of the anomaly detection pipeline, including training regularization, prompt engineering, anomaly scoring mechanisms, backbone adaptation, and direct anomaly score prediction. Early experiments focus on evaluating each modification independently, while later experiments combine the strongest components to determine whether their improvements are complementary. All experiments are evaluated under identical dataset splits and few-shot settings to ensure fair comparisons against the baseline MVFA-AD implementation.

Baseline and Optimization Experiments

Experiment 1 — Adapter Regularization + Better Optimization

Compared to the baseline configuration, several targeted training modifications were introduced while keeping the overall pipeline unchanged. Specifically, weight decay with a default value of $1e-4$ was applied to the adapter parameters to improve regularization and reduce overfitting. Additionally, a cosine annealing learning rate schedule was adopted to provide smoother learning rate decay during training, potentially leading to more stable convergence. Gradient clipping with a maximum norm of 1.0 was also implemented to prevent exploding gradients and maintain training stability. Aside from these adjustments, all other components—including the model architecture, dataset, anomaly scoring method, and plotting procedures—remain identical to the original baseline setup, ensuring that any observed performance differences can be attributed solely to these training modifications.

Prompt Engineering Experiments

Experiment 2 — Prompt / Text-Side Improvement

In this experiment, an additional modification introduces a medical-domain prompt ensembling tailored specifically for chest X-ray analysis compared to the baseline. Instead of relying on generic prompts, the model uses Chest X-Ray–specific textual descriptions that capture both normal and abnormal findings. Furthermore, the abnormal prompts are designed at multiple levels of granularity, combining disease-specific descriptions (e.g., pneumonia-related abnormalities) with generic abnormality prompts that capture broader pathological patterns. This prompt diversification enables the model to better align visual features with clinically meaningful textual representations, thereby improving the robustness of anomaly detection by capturing both specific disease indicators and general abnormal characteristics in chest radiographs.

Anomaly Scoring Experiments

Experiment 3 — Better Anomaly Scoring Head

This experiment focuses on changing how anomaly scoring is performed, while keeping the training procedure unchanged. Specifically, Mahalanobis distance scoring was applied by fitting a Gaussian distribution to the normal feature representations and measuring deviations at test time. In addition, an energy-based zero-shot scoring method using logsumexp to estimate anomaly likelihood from the model logits. A learned temperature scaling parameter (τ) was also introduced to calibrate the logits before computing the final scores. Importantly, the training loop remains identical to the baseline, with changes applied only to the test-time evaluation and post-training scoring functions.

Vision Backbone Adaptation Experiments

Experiment 4 — LoRA Vision-Side Adaptation

In this experiment, we introduce LoRA applied to the final ViT layers. Specifically, LoRA is inserted into the last two transformer blocks (blocks 22 and 23), with a default rank of $r = 4$ and $\alpha = 4$. The LoRA parameters are trained jointly with the adapter parameters and included in the optimizer. All other components, including the dataset, loss functions, testing procedure, and anomaly scoring, remain identical to the baseline configuration to ensure a fair comparison.

Combined Architecture Experiments

Experiment 5 — Regularized Scoring Head + Medical Prompts

This experiment combines the strongest components from Experiments 1, 2, and 3 into a unified configuration. In Experiment 1, the training procedure incorporates weight decay, a cosine annealing learning rate schedule, and gradient clipping to improve optimization stability. From Experiment 2, the text encoder utilizes medical-domain prompt ensembling with chest X-ray–specific normal and abnormal descriptions. From Experiment 3, the evaluation stage employs a fused anomaly scoring

strategy that combines Mahalanobis distance, energy-based scoring, and cosine similarity. The training loop follows the regularization scheme from Experiment 1, the text features are generated using the medical prompts from Experiment 2, and the testing pipeline applies the multi-score fusion from Experiment 3.

Experiment 6 — LoRA + Medical Prompts + Regularization

This configuration integrates the most effective elements from Experiments 1, 2, and 4. The training process adopts the regularization and optimization strategies from Experiment 1, including weight decay, a cosine annealing learning rate schedule, and gradient clipping to stabilize training. From Experiment 2, the model incorporates medical-domain prompt ensembling using chest X-ray-specific descriptions of normal and abnormal findings to better align the text representations with the medical imaging domain. In addition, Experiment 4 provides parameter-efficient fine-tuning via LoRA applied to the final ViT blocks, enabling targeted adaptation of the backbone while keeping the majority of the pretrained weights unchanged.

Experiment 7 — MedPrompts + Fast GPU Scoring

The underlying hypothesis is that the additional regularization introduced in Experiment 5 constrained the adapters too strongly, limiting their capacity to accurately model the complex Mahalanobis feature distribution. By removing that regularization, the model regains the raw fitting flexibility observed in Experiment 3 while still benefiting from the improved textual alignment introduced in Experiment 2 through medical-domain prompts.

Experiment 8 — MedPrompts + Fast GPU Scoring + LoRA

The hypothesis behind this configuration is a “Grand Fusion” approach that simultaneously adapts all major components of the system. The text branch is improved through medical-domain prompt ensembling tailored to chest X-ray descriptions, the vision backbone is adapted using LoRA applied to the final transformer blocks, and the anomaly scoring mechanism combines Mahalanobis and energy-based metrics. By removing the strict regularization introduced in Experiment 1, the model is expected to regain flexibility in the few-shot regime, enabling the combined adaptations across text, vision, and scoring components to work together more effectively and improve low-shot performance.

Experiment 9 — MedPrompts + LoRA

As for experiment 9, we hypothesize that Experiment 6, which used the same combination with strong regularization, performed poorly in the 2-shot setting because the regularization limited the model’s flexibility. By removing these constraints, the model should recover the low-shot performance observed in Experiment 4 while retaining the higher-shot improvements from combining medical prompts with LoRA.

Backbone Replacement Experiment

Experiment 10 — BiomedCLIP Backbone Swap

Compared to the baseline, we replace OpenAI’s general-purpose ViT-L/14 CLIP with Microsoft’s domain-adapted BiomedCLIP model. This change also requires modifying the tokenization pipeline to support the custom BiomedCLIP tokenizer. The hypothesis is that, because BiomedCLIP is pretrained on PubMed text and medical imaging data, its learned feature representations should more naturally separate normal from abnormal chest X-rays than those of the general-domain CLIP model.

Direct Visual Anomaly Prediction

Experiment 11 — Anomaly Score Prediction

In this experiment, the anomaly scoring pipeline was modified by removing text features and by introducing an MLP that directly predicts the anomaly score from image features. This change removes reliance on text prompts and prompt

engineering, resulting in a simpler scoring mechanism. By allowing the model to learn anomaly patterns directly from visual representations, the approach focuses solely on image information, leading to improved performance in our experiments.

Fusion and Calibration Experiments

Experiment 12 — Exp 8 + Learned Fusion (drop Energy) + Diagonal Mahalanobis

This experiment builds on Experiment 8 (Medical Prompts + Score + LoRA without regularization) with several refinements to improve stability. Energy-based scoring was removed because it introduced noise at mid-shot settings. The full-covariance Mahalanobis distance was replaced with a diagonal variant to provide more stable behavior across all shot counts. Finally, a learned fusion mechanism based on logistic regression was introduced to combine cosine similarity and Mahalanobis scores, enabling the model to adaptively weight both signals for more reliable anomaly detection.

Experiment 13 — Additive LoRA & Z-score Results

This experiment extends the configuration of Experiment 12 with two methodological adjustments. Z-score normalization is used for score calibration rather than min-max normalization, and the LoRA implementation uses the standard additive formulation across the Q, K, and V projection layers rather than cloning and modifying the attention weights. All other components, including medical-domain prompt ensembling, diagonal Mahalanobis scoring, and the learned fusion mechanism, remain unchanged.

Experiment 14 — Exp5 + DiagMahal + 3-Score Learned Fusion

This experiment builds on the Medical Prompts + Score configuration of Experiment 5 with modifications to the scoring pipeline. The full-covariance Mahalanobis distance is replaced with a diagonal Mahalanobis variant to improve numerical stability. The resulting scores are then standardized using Z-score calibration and fused through a Logistic Regression model that combines cosine similarity, diagonal Mahalanobis distance, and energy-based scores.

Test-Time Calibration Experiments

Experiment 15 — Exp4 LoRA + Test-Time Calibrated 3-Score Fusion + Temperature Sweep

The training procedure follows the same configuration as Experiment 4, which applies LoRA-based adaptation on the vision encoder. Several enhancements are introduced during inference. A full-covariance Mahalanobis distance with shrinkage ($\alpha = 0.1$) is used to improve stability in higher-shot settings. The anomaly score is derived from a calibrated combination of cosine similarity, energy score, and Mahalanobis distance. Normalization statistics are computed exclusively from the calibration (support) set to prevent any test leakage. The final anomaly score is produced using a Logistic Regression fusion model ($C = 0.1$, balanced class weights), with temperature scaling evaluated at inference using values $\{0.7, 1.0, 1.3, 1.6\}$. Additionally, medical-domain prompt ensembling is applied to incorporate clinically relevant textual context.

Experiment 16 — Exp15 + High-Capacity LoRA (rank=8, 3 blocks)

This experiment follows the same configuration as Exp15, but with increased LoRA adaptation capacity. The LoRA rank is raised to 8 (previously 4), and the adapted transformer blocks are expanded to include the last three layers ($[-3, -2, -1]$) rather than only the final two. In addition, the LoRA scaling factor (α) is increased from 1.0 to 8.0 to allow stronger updates via the low-rank adapters.

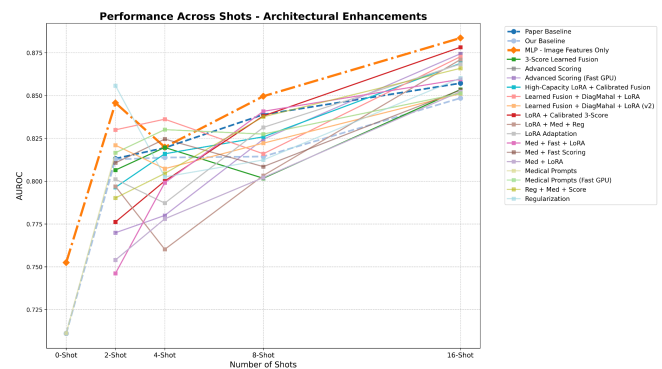


Figure 2. Comparative results of different experiments

3.3.2 Combined Results

The combined results matrix reveals a clear architectural scaling law: no single modification optimally generalizes across all data constraints. At the extreme low-data limit (2-shot), simple adapter regularization (Experiment 1) decisively outperforms all complex variants, achieving a peak AUROC of 0.8558 by physically preventing the adapter from overfitting the tiny support set. As the target images increase slightly (4-shot), our learned fusion architecture (Experiment 12) takes the lead (0.8362) due to its ability to dynamically calibrate scoring thresholds. However, in the zero-shot baseline and highest available data regimes (8-shot and 16-shot), the direct anomaly score Prediction MLP (Experiment 11) aggressively dominates, yielding the absolute highest overall test performance of 0.8836 at 16-shots. This progression indicates that while heavy mathematical regularization is mandatory to prevent mode collapse under severe constraints (≤ 2 shots), discarding multimodal language embeddings entirely in favor of direct visual regression (MLP) maximizes anomaly discrimination once sufficient pathological variance is available in the support set (≥ 8 shots).

Entries marked N/A correspond to configurations that are not applicable to the zero-shot setting or require support-set fitting procedures that cannot be performed without labeled target examples.

Model	zero-shot	2-shot	4-shot	8-shot	16-shot
Paper Baseline	0.7111	0.8132	0.8195	0.8389	0.8572
Our Baseline	0.7112	0.8127	0.8138	0.8144	0.8485
Exp1 Regularization	N/A	0.8558	0.8025	0.8124	0.8602
Exp2 Medical prompt ensemble	0.7112	0.8166	0.8301	0.8451	0.8513
Exp3 Enhanced scoring	N/A	0.7499	0.7800	0.8245	0.8744
Exp4 LoRA Vision Adaptation	N/A	0.8011	0.7872	0.8314	0.8684
Exp5 Combined(Reg + prompt + score)	N/A	0.7902	0.8045	0.8376	0.8659
Exp6 Combined(Reg + prompt +LoRA)	N/A	0.7969	0.7602	0.8033	0.8710
Exp7 Combined(Prompt + score)	N/A	0.8107	0.8246	0.8085	0.8532
Exp8 Combined(Prompt + score + LoRA)	N/A	0.7462	0.7991	0.8408	0.8595
Exp9 Combined(Prompt + LoRA)	N/A	0.7540	0.7780	0.8021	0.8523
Exp10 BiomedCLIP	0.6000	0.8234	0.8061	0.8159	0.8440
Exp11 MLP	0.7525	0.8457	0.8199	0.8496	0.8836
Exp12 Learned Fusion	N/A	0.8300	0.8362	0.8160	0.8727
Exp13 Additive LoRA & Z-score	N/A	0.8210	0.8073	0.822	0.8511
Exp14 3-score fusion	N/A	0.8065	0.8197	0.8017	0.8535
Exp15 Calibrated fusion + temp sweep	N/A	0.7762	0.8001	0.8381	0.8782
Exp16 high capacity LoRA	N/A	0.7964	0.8160	0.8259	0.8688

Table 1: AUROC results across all experimental configurations under different few-shot regimes.

4. Discussion & Takeaways

4.1 Decisions made

Several design decisions were made throughout the project to isolate the impact of different architectural modifications. Model architecture complexity was separated from data scaling effects by independently evaluating components such as regularization, prompt engineering, anomaly scoring strategies, and backbone adaptation. Experiments were first designed to test each modification individually, followed by combined configurations to determine whether their improvements were complementary. A strong emphasis was placed on few-shot stability, motivating targeted experiments on regularization (Experiment 1), medical-domain prompt engineering (Experiment 2), and improved scoring metrics (Experiment 3). Additionally, an exploratory analysis of the RSNA dataset distribution and its labels was conducted early in the project to identify potential failure modes and guide subsequent experimental design.

4.2 Difficulties faced

Several technical challenges emerged during the development and evaluation of the anomaly detection pipeline. Covariance matrix instability occurred when estimating Mahalanobis distances with extremely small support sets (2-shot or 4-shot), as the high dimensionality of visual embeddings led to poorly conditioned covariance matrices and unstable inversions. Additional difficulties arose during score fusion because cosine similarity, Mahalanobis distance, and energy scores operate on different numerical scales, requiring careful normalization and calibration. Furthermore, integrating multiple architectural modifications simultaneously often introduced training instability, particularly under extreme few-shot conditions.

4.3 Things that worked

Several modifications consistently improved model stability and detection performance. Medical prompt engineering (Experiment 2) improved stability by aligning text embeddings with clinically meaningful descriptions of chest X-ray abnormalities. Diagonal Mahalanobis scoring provided a more stable alternative to full covariance estimation, particularly in low-shot regimes. Multi-score fusion architectures further enhanced anomaly detection by combining complementary scoring mechanisms. Finally, the MLP-based anomaly score prediction architecture (Experiment 11) achieved the strongest overall performance, reaching an AUROC of 0.8836 at 16 shots, demonstrating that direct visual anomaly regression can outperform similarity-based scoring when sufficient training examples are available.

4.4 Things that didn't work

Some experimental modifications did not perform as expected. Replacing the backbone with BiomedCLIP (Experiment 10) resulted in worse performance than the standard CLIP backbone, likely because the altered visual token structure reduced the adapter's ability to capture localized abnormal regions. Additionally, combining too many mechanisms simultaneously, such as regularization, prompt engineering, and multiple scoring heads, reduced robustness and introduced instability. Prompt learning using context optimization (CoOp) also failed to converge effectively in the medical domain, as the learnable prompt tokens drifted toward trivial semantic representations that did not meaningfully distinguish normal and abnormal chest X-rays.

5. Conclusion

Across the experiment matrices evaluated on the RSNA benchmark, results consistently indicate that structural adaptation mechanisms contribute more to few-shot medical anomaly detection performance than model scaling alone. Medical prompting and heavy regularization prove most effective at extremely limited sample sizes (2-shot), while enhanced metric scaling yields stronger gains as labeled data increases (16-shot). A limitation of this work is that experiments were conducted on a single dataset due to computational constraints, which may restrict the generalizability of the findings. Future work should evaluate the proposed architectures across additional medical imaging benchmarks and investigate hybrid approaches that combine visual representation learning with language-guided feature alignment.

6. Disclaimer

ChatGPT was used to refine the language and fix the grammar of the report. Claude was used in code generation.

7. References

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